DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-200-02 — Internal inspection of vertical vessel V-101

Area	A200 Vessels and column	**Task**	VI Internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**V-101** — V-101 vertical vessel 72" ID × 216" T/T — E 33'-6" N -25'-0" EL 100'-8-1/4"		
Equipment	V-101	**Lines**	4"-P300-P-1005, 6"-P300-P-1006, 1"-P150-V-6001, 1"-P150-D-6002, 1"-P150-N-3011, 3/4"-P300-I-6003, 3/4"-P300-I-6004, 3/4"-P300-SP-6005
Isometrics	ISO-4"-P300-P-1005, ISO-6"-P300-P-1006, ISO-1"-P150-V-6001, ISO-1"-P150-D-6002, ISO-1"-P150-N-3011, ISO-3/4"-P300-I-6003, ISO-3/4"-P300-I-6004, ISO-3/4"-P300-SP-6005		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the vessel.

Scope of work

1. Isolate V-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 14 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: V-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
V-101	equipment	V-101 vertical vessel 72" ID × 216" T/T		E 33'-6" N -25'-0" EL 100'-8-1/4"
4"-P300-P-1005/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 33'-6" N -20'-8-3/16" EL 121'-0"
V-101-N1	nozzle	Nozzle V-101-N1 (inlet, 4" P300) on V-101		E 33'-6" N -21'-3-15/16" EL 121'-0"
6"-P300-P-1006/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1006	E 33'-6" N -20'-8-9/16" EL 109'-0"
V-101-N2	nozzle	Nozzle V-101-N2 (outlet, 6" P300) on V-101		E 33'-6" N -21'-3-11/16" EL 109'-0"
1"-P150-V-6001/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6001	E 33'-6" N -25'-0" EL 128'-8-3/4"
V-101-N3	nozzle	Nozzle V-101-N3 (vent, 1" P150) on V-101		E 33'-6" N -25'-0" EL 128'-1-15/16"
1"-P150-D-6002/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6002	E 37'-6-1/2" N -25'-0" EL 108'-6"
V-101-N4	nozzle	Nozzle V-101-N4 (drain, 1" P150) on V-101		E 36'-11-11/16" N -25'-0" EL 108'-6"
1"-P150-N-3011/B2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-N-3011	E 37'-6-1/2" N -25'-0" EL 120'-6"
V-101-N5	nozzle	Nozzle V-101-N5 (nitrogen, 1" P150) on V-101		E 36'-11-11/16" N -25'-0" EL 120'-6"
3/4"-P300-I-6003/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6003	E 37'-6-1/4" N -25'-0" EL 112'-6"
V-101-N6	nozzle	Nozzle V-101-N6 (instrument, 0.75" P300) on V-101		E 36'-11-9/16" N -25'-0" EL 112'-6"
3/4"-P300-I-6004/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6004	E 37'-6-1/4" N -25'-0" EL 118'-0"
V-101-N7	nozzle	Nozzle V-101-N7 (instrument, 0.75" P300) on V-101		E 36'-11-9/16" N -25'-0" EL 118'-0"
3/4"-P300-SP-6005/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-SP-6005	E 37'-6-1/4" N -25'-0" EL 110'-6"
V-101-N8	nozzle	Nozzle V-101-N8 (sample, 0.75" P300) on V-101		E 36'-11-9/16" N -25'-0" EL 110'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1005/B2	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B2) on V-101-N1	E 33'-6" N -20'-8-3/16" EL 121'-0"		
2	blind	V-101-N1	4"-P300-P-1005	Fit a blind between 4"-P300-P-1005/B2 and the nozzle flange V-101-N1	E 33'-6" N -21'-3-15/16" EL 121'-0"		
3	valve	6"-P300-P-1006/A2	6"-P300-P-1006	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1006/A2) on V-101-N2	E 33'-6" N -20'-8-9/16" EL 109'-0"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
4	blind	V-101-N2	6"-P300-P-1006	Fit a blind between 6"-P300-P-1006/A2 and the nozzle flange V-101-N2	E 33'-6" N -21'-3-11/16" EL 109'-0"		
5	valve	1"-P150-V-6001/2	1"-P150-V-6001	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6001/2) on V-101-N3	E 33'-6" N -25'-0" EL 128'-8-3/4"		
6	blind	V-101-N3	1"-P150-V-6001	Fit a blind between 1"-P150-V-6001/2 and the nozzle flange V-101-N3	E 33'-6" N -25'-0" EL 128'-1-15/16"		
7	valve	1"-P150-D-6002/2	1"-P150-D-6002	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6002/2) on V-101-N4	E 37'-6-1/2" N -25'-0" EL 108'-6"		
8	blind	V-101-N4	1"-P150-D-6002	Fit a blind between 1"-P150-D-6002/2 and the nozzle flange V-101-N4	E 36'-11-11/16" N -25'-0" EL 108'-6"		
9	valve	1"-P150-N-3011/B2	1"-P150-N-3011	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3011/B2) on V-101-N5	E 37'-6-1/2" N -25'-0" EL 120'-6"		
10	blind	V-101-N5	1"-P150-N-3011	Fit a blind between 1"-P150-N-3011/B2 and the nozzle flange V-101-N5	E 36'-11-11/16" N -25'-0" EL 120'-6"		
11	valve	3/4"-P300-I-6003/2	3/4"-P300-I-6003	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6003/2) on V-101-N6	E 37'-6-1/4" N -25'-0" EL 112'-6"		
12	blind	V-101-N6	3/4"-P300-I-6003	Fit a blind between 3/4"-P300-I-6003/2 and the nozzle flange V-101-N6	E 36'-11-9/16" N -25'-0" EL 112'-6"		
13	valve	3/4"-P300-I-6004/2	3/4"-P300-I-6004	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6004/2) on V-101-N7	E 37'-6-1/4" N -25'-0" EL 118'-0"		
14	blind	V-101-N7	3/4"-P300-I-6004	Fit a blind between 3/4"-P300-I-6004/2 and the nozzle flange V-101-N7	E 36'-11-9/16" N -25'-0" EL 118'-0"		
15	valve	3/4"-P300-SP-6005/2	3/4"-P300-SP-6005	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6005/2) on V-101-N8	E 37'-6-1/4" N -25'-0" EL 110'-6"		
16	blind	V-101-N8	3/4"-P300-SP-6005	Fit a blind between 3/4"-P300-SP-6005/2 and the nozzle flange V-101-N8	E 36'-11-9/16" N -25'-0" EL 110'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-200-02-I1	Internal visual examination of the shell, heads and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-200-02-I2	Ultrasonic thickness at the 14 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-200-02-TML01	V-101	Nozzle neck V-101-N1	E 33'-6" N -21'-3-15/16" EL 121'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML02	V-101	Nozzle neck V-101-N2	E 33'-6" N -21'-3-11/16" EL 109'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML03	V-101	Nozzle neck V-101-N3	E 33'-6" N -25'-0" EL 128'-1-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML04	V-101	Nozzle neck V-101-N4	E 36'-11-11/16" N -25'-0" EL 108'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML05	V-101	Nozzle neck V-101-N5	E 36'-11-11/16" N -25'-0" EL 120'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML06	V-101	Nozzle neck V-101-N6	E 36'-11-9/16" N -25'-0" EL 112'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML07	V-101	Nozzle neck V-101-N7	E 36'-11-9/16" N -25'-0" EL 118'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML08	V-101	Nozzle neck V-101-N8	E 36'-11-9/16" N -25'-0" EL 110'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-02-TML09	V-101	Bottom head centre	E 33'-6" N -25'-0" EL 105'-2"		centre: ____			
WO-200-02-TML10	V-101	Top head centre	E 33'-6" N -25'-0" EL 126'-2-1/2"		centre: ____			
WO-200-02-TML11	V-101	Shell at 0.000" above the lower tangent	E 33'-6" N -25'-0" EL 106'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-02-TML12	V-101	Shell at 6'-0" above the lower tangent	E 33'-6" N -25'-0" EL 112'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-02-TML13	V-101	Shell at 12'-0" above the lower tangent	E 33'-6" N -25'-0" EL 118'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-02-TML14	V-101	Shell at 18'-0" above the lower tangent	E 33'-6" N -25'-0" EL 124'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

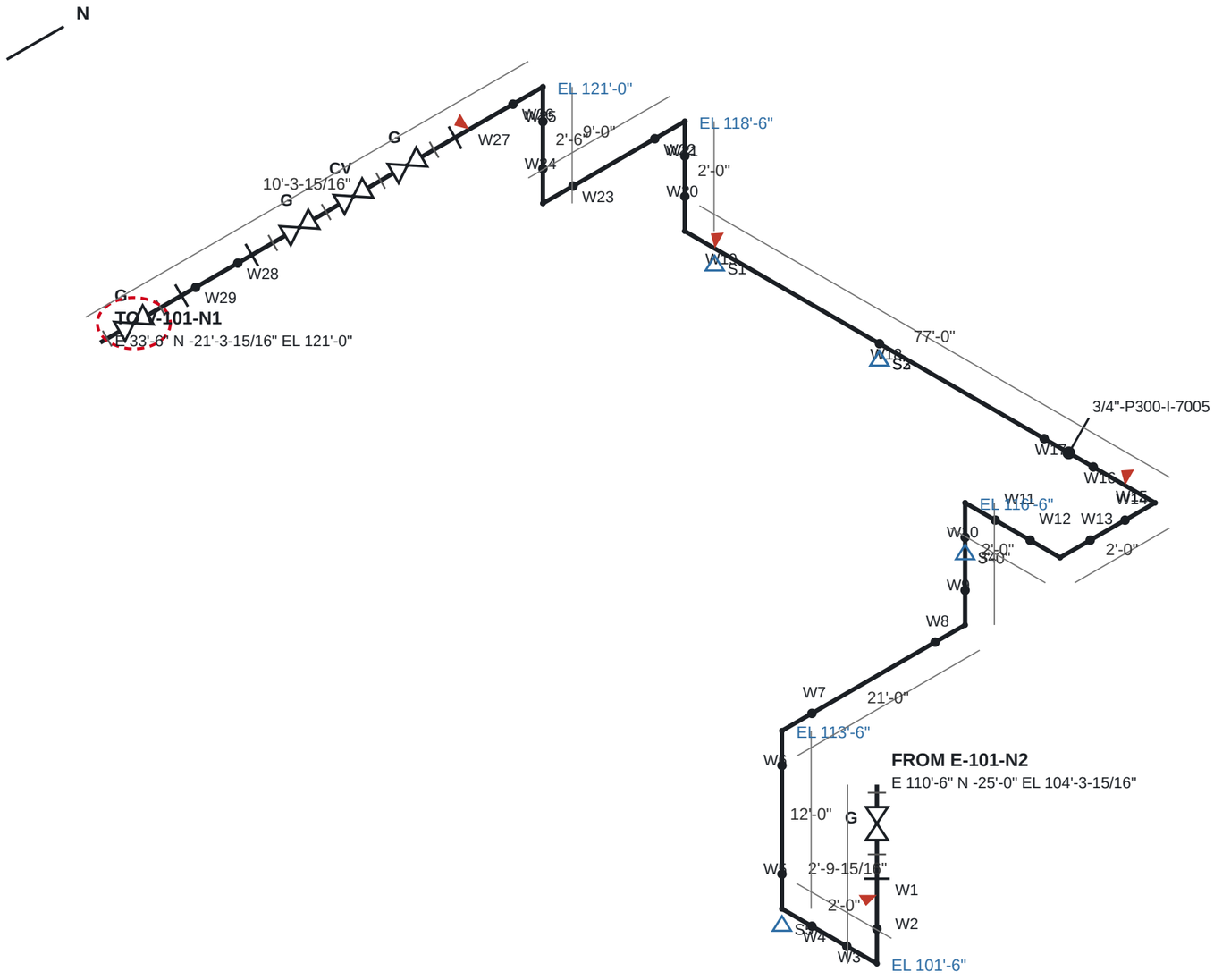
Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 14 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1005 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	8
2	VALVE, GATE, WEDGE, OS&Y 4" P300	4
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	8
4	FLANGE WELD-NECK 4" P300 RAISED FACE	4
5	PIPE 4" P300 BEVELLED ENDS	124'-6-1/2"
6	ELBOW 90° R = 5/3 OD 4" P300	11
7	TEE REDUCING 4" × 3/4" P300	1
8	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1

WELDS 29 (4 FIELD) · SUPPORTS 5 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F W16 W17 W18 W19F W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

4"-P300-P-1005

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-101-N2 → V-101-N1
PIPE 124'-6-1/2" · WEIGHT 1974 lb

DWG ISO-4"-P300-P-1005 REV 0 WO-200-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-6"-P300-P-1006

ISOMETRIC — 6"-P300-P-1006 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	3
5	PIPE 6" P300 BEVELLED ENDS	175'-11-5/8"
6	ELBOW 90° R = 5/3 OD 6" P300	8
7	TEE REDUCING 6" × 3/4" P300	2
8	TEE EQUAL 6" P300	1
9	REDUCER CONCENTRIC 6" × 4" P300	1
10	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1
11	FLANGE WELD-NECK 4" P300 RAISED FACE	1
12	GASKET 4" P300 COMPRESSED SHEET	1

WELDS 29 (4 FIELD) · SUPPORTS 8 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9F W10
W11 W12 W13 W14 W15 W16 W17F W18 W19 W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

6"-P300-P-1006

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
V-101-N2 → P-102A-S
PIPE 175'-11-5/8" · WEIGHT 3070 lb

DWG ISO-6"-P300-P-1006 REV 0 WO-200-02

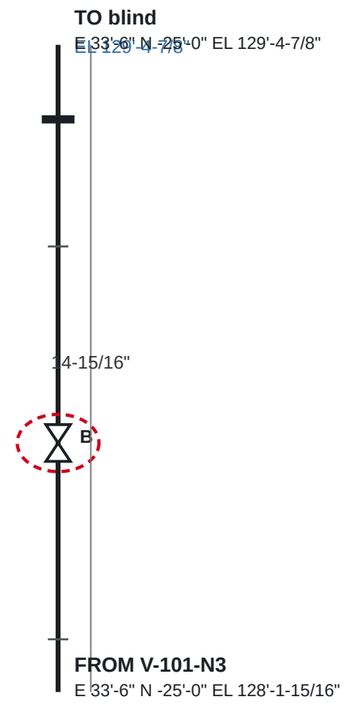
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Isometric ISO-1"-P150-V-6001

ISOMETRIC — 1"-P150-V-6001 — Internal inspection of vertical vessel V-101

N



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6001

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
V-101-N3 → blind
PIPE 0.000" · WEIGHT 39 lb

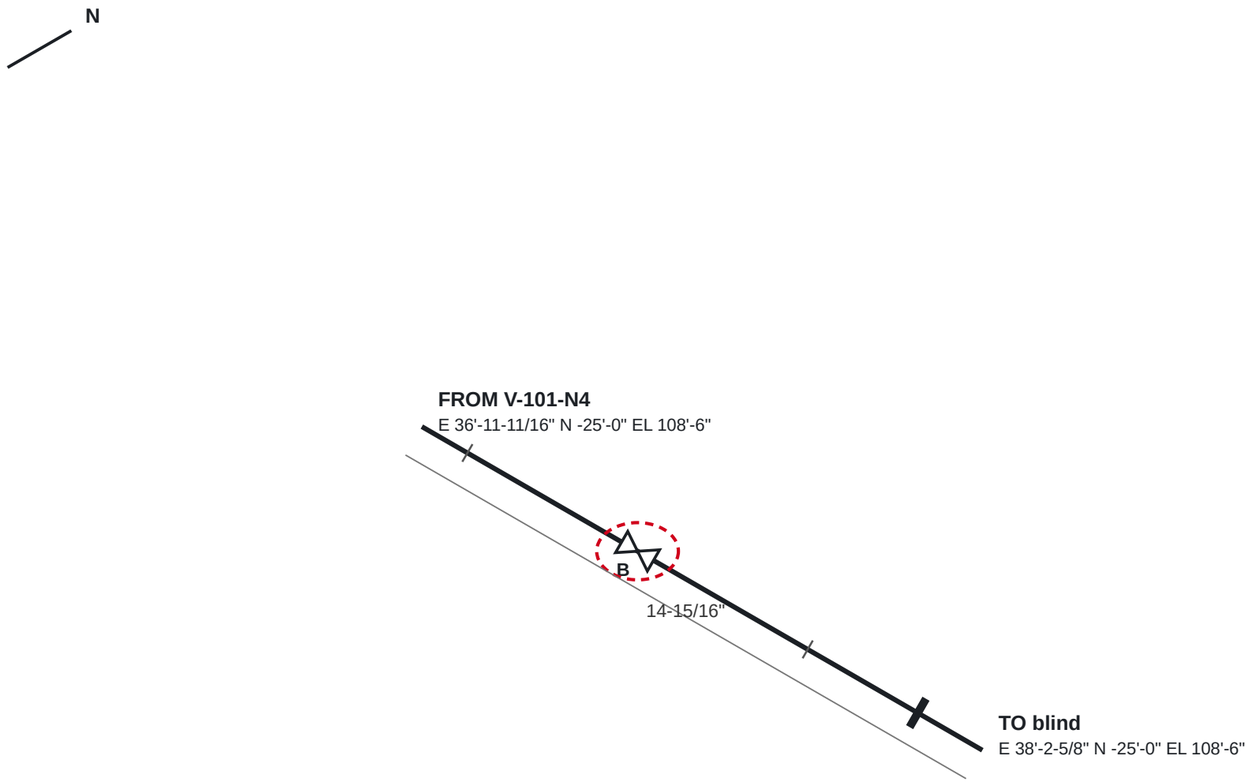
DWG ISO-1"-P150-V-6001 REV 0 WO-200-02

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Isometric ISO-1"-P150-D-6002

ISOMETRIC — 1"-P150-D-6002 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6002

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
V-101-N4 → blind
PIPE 0.000" · WEIGHT 39 lb

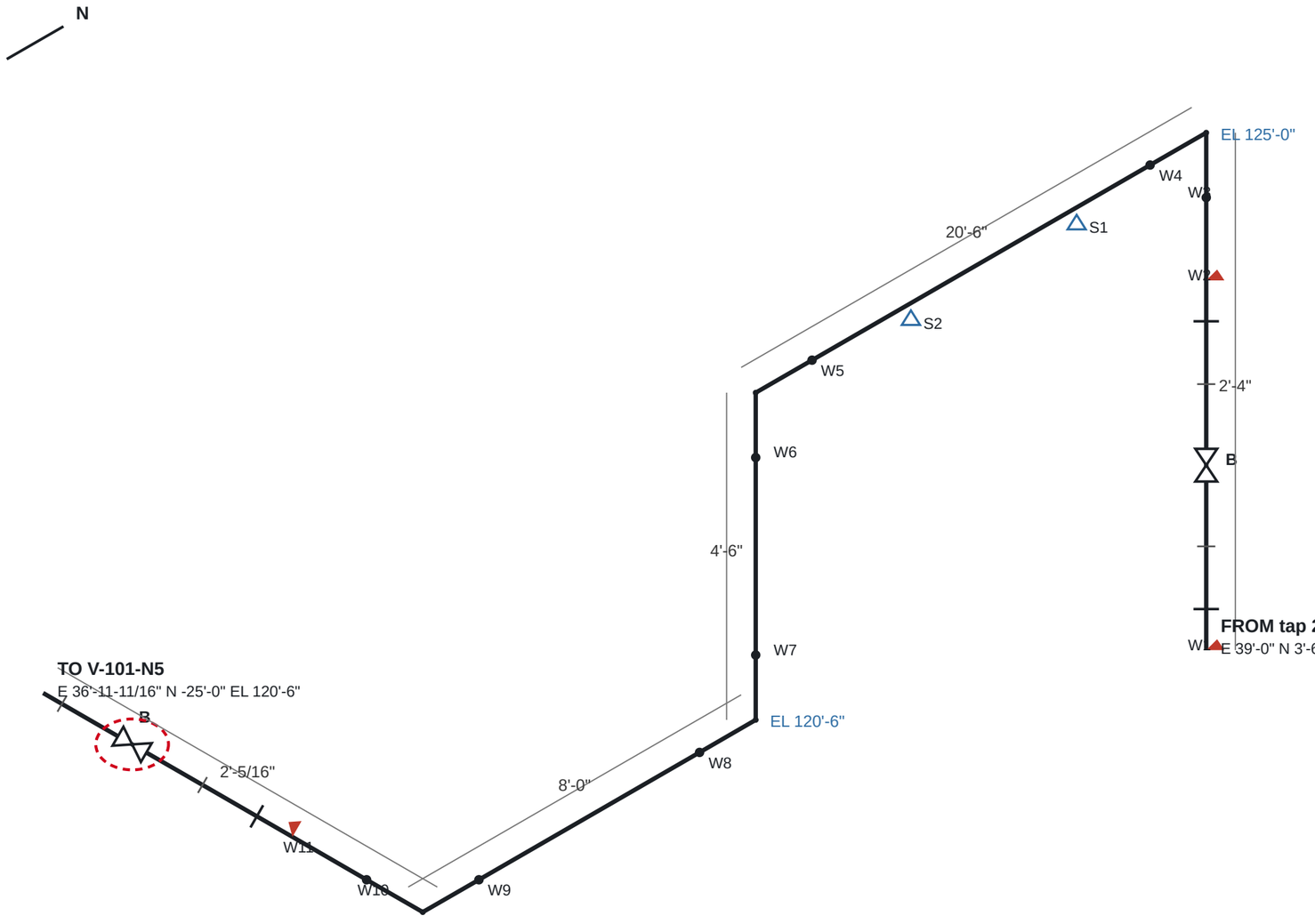
DWG ISO-1"-P150-D-6002 REV 0 WO-200-02

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Isometric ISO-1"-P150-N-3011

ISOMETRIC — 1"-P150-N-3011 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P150 RAISED FACE	3
2	GASKET 1" P150 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P150	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	4
5	PIPE 1" P150 BEVELLED ENDS	32'-11-3/16"
6	ELBOW 90° R = 5/3 OD 1" P150	4

WELDS 11 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

1"-P150-N-3011

CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → V-101-N5
PIPE 32'-11-3/16" · WEIGHT 114 lb

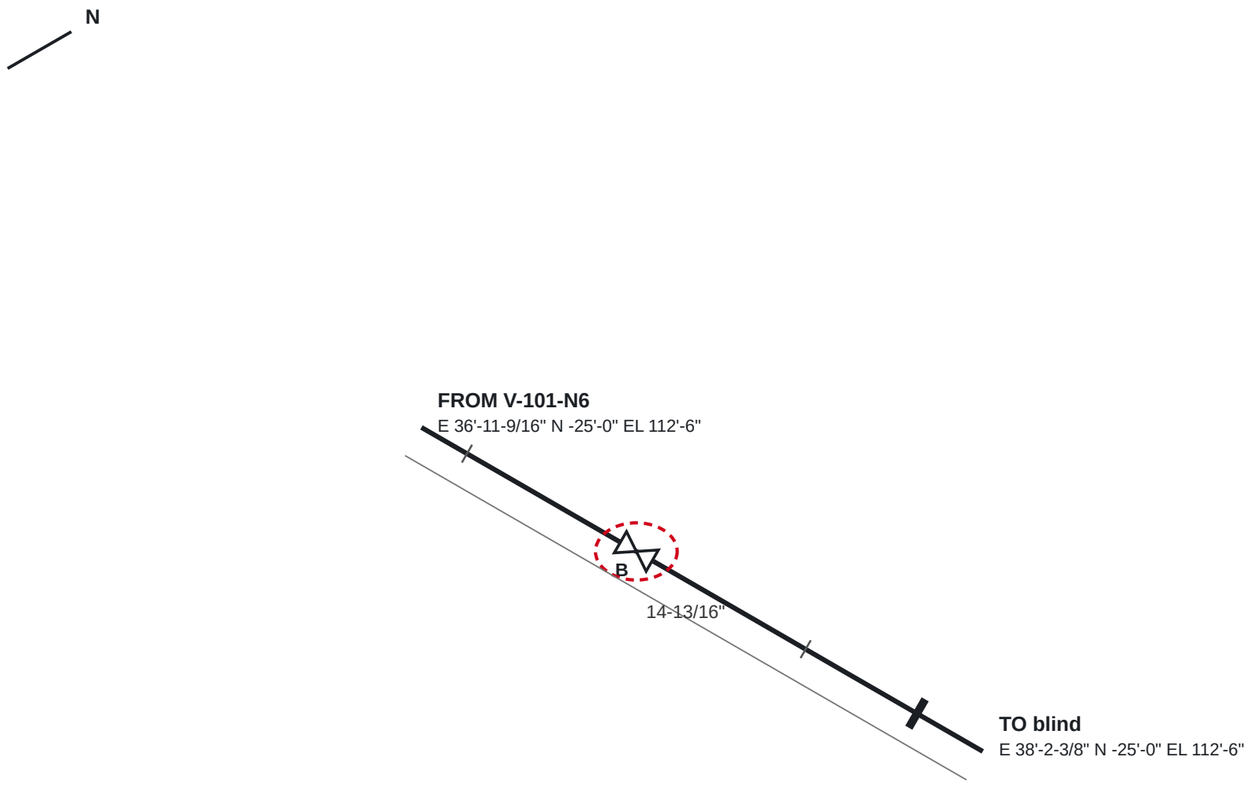
DWG ISO-1"-P150-N-3011 REV 0 WO-200-02

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Isometric ISO-3/4"-P300-I-6003

ISOMETRIC — 3/4"-P300-I-6003 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6003

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
V-101-N6 → blind
PIPE 0.000" · WEIGHT 32 lb

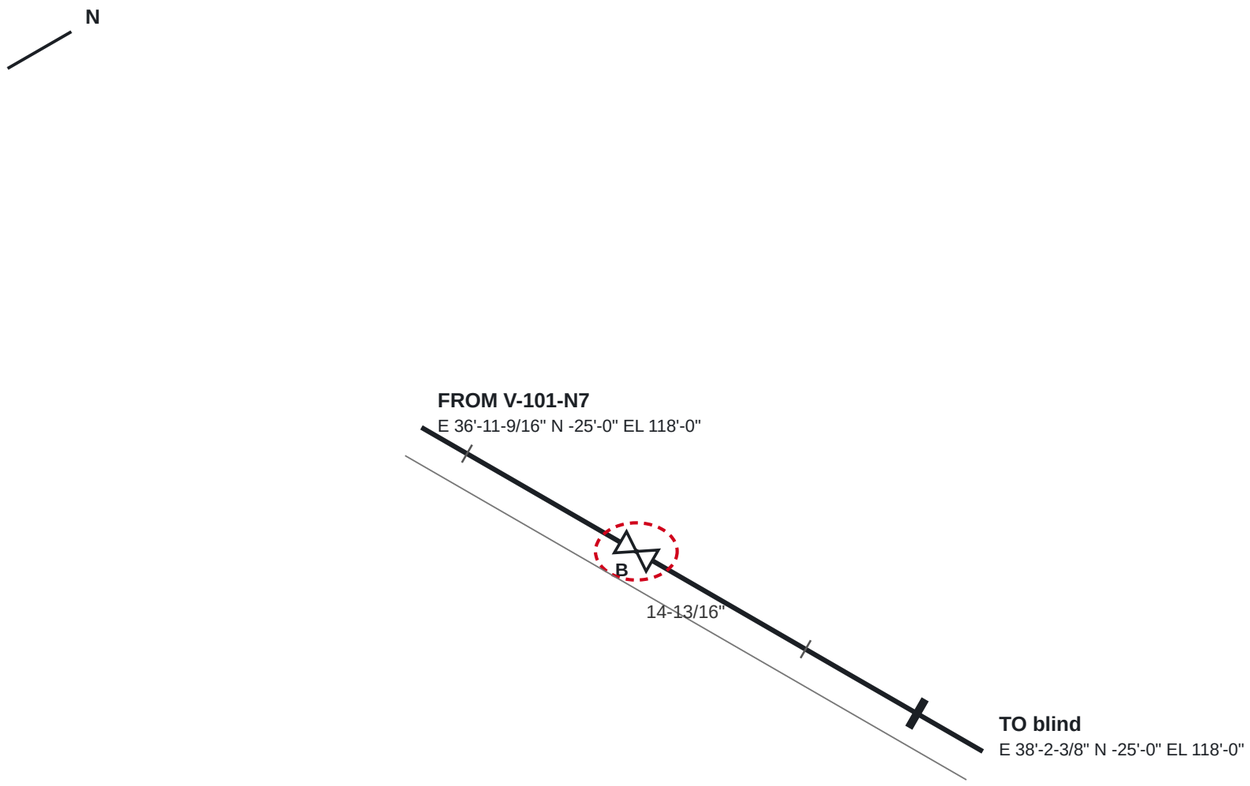
DWG ISO-3/4"-P300-I-6003 REV 0 WO-200-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-3/4"-P300-I-6004

ISOMETRIC — 3/4"-P300-I-6004 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6004

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
V-101-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

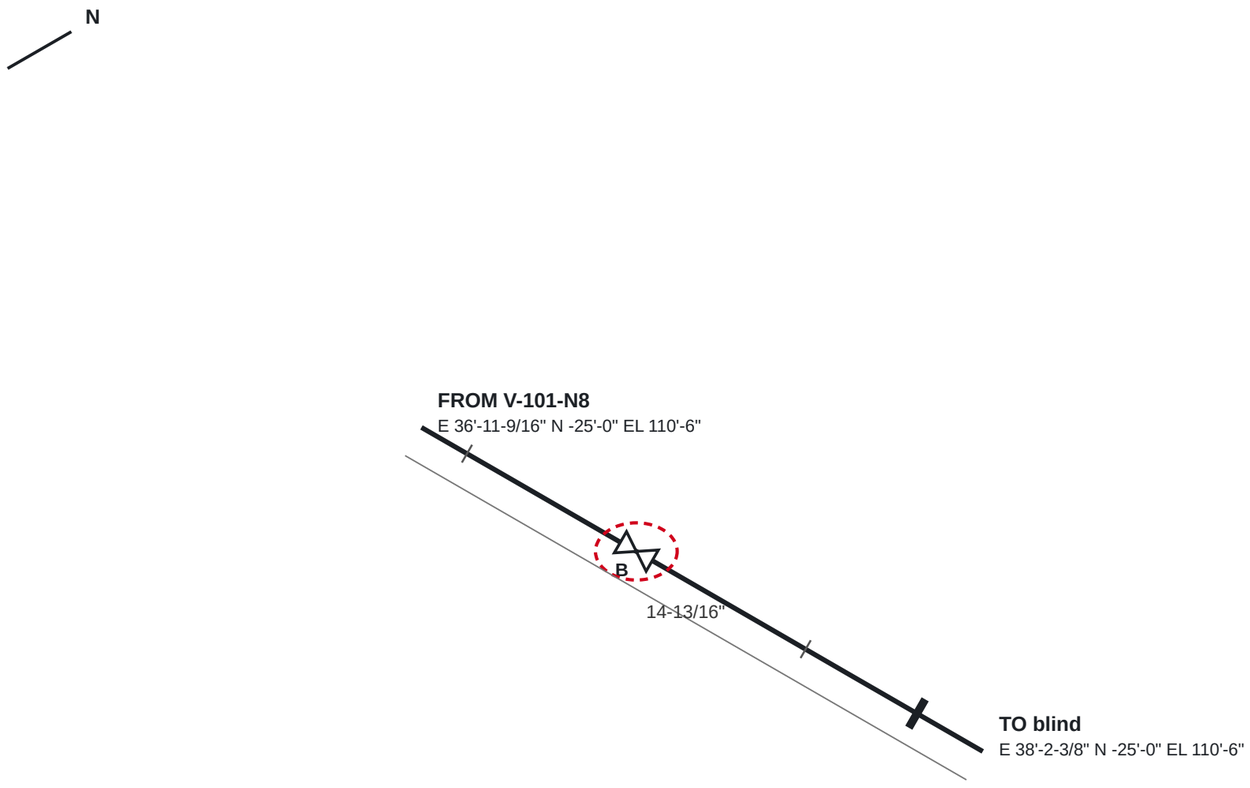
DWG ISO-3/4"-P300-I-6004 REV 0 WO-200-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-SP-6005

ISOMETRIC — 3/4"-P300-SP-6005 — Internal inspection of vertical vessel V-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-SP-6005

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
V-101-N8 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P300-SP-6005 REV 0 WO-200-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE